MACHINE LEARNING

ASSIGNMENT 03

Q1. Define Supervised Learning. Explain the key difference between supervised learning and unsupervised learning with one example of each.

Supervised Learning is a machine learning technique in which a model is trained using labelled data. The data contains input features and their correct outputs, allowing the model to learn patterns and make predictions on new data.

Difference between Supervised and Unsupervised Learning:

Basis	**Supervised Learning**	**Unsupervised Learning**
Data	Uses labelled data.	Uses unlabelled data.
Goal	Predicts output or target values.	Finds hidden patterns or groups.
Example	Predicting house prices using past data.	Grouping customers based on shopping behaviour.
Algorithms	Linear Regression, Decision Tree.	K-Means Clustering, PCA.

Example of Supervised Learning: Predicting house prices using features such as area, location, and number of rooms. The model learns from previously sold houses with known prices.

Example of Unsupervised Learning: Grouping customers according to their purchasing behaviour using the K-Means clustering algorithm.

Thus, supervised learning predicts known target values, whereas unsupervised learning discovers patterns in data without predefined labels.

Q2. What are the two main types of supervised learning problems? Clearly distinguish between them and give one real-world example for each type.

The two main types of supervised learning problems are classification and regression.

1. Classification

Classification is used when the output belongs to a fixed category or class. The model learns from labelled examples and predicts the category of new data.

- Output: Discrete categories or classes.
- Algorithms: Logistic Regression, Decision Tree, and Support Vector Machine.
- Example: An email classification system identifies whether an email is spam or not spam.

2. Regression

Regression is used when the output is a continuous numerical value. It establishes a relationship between input variables and a numerical target.

- Output: Continuous numerical values.
- Algorithms: Linear Regression, Decision Tree Regressor, and Random Forest Regressor.
- Example: Predicting the selling price of a house based on its size, location, and number of rooms.

Q3. Explain the concept of Linear Regression. Write the equation of a simple linear regression model and define the terms. What is the role of the residual/error term?

Linear Regression is a supervised learning algorithm used to predict a continuous numerical value by establishing a linear relationship between an independent variable and a dependent variable.

For example, it can predict a student's marks based on the number of hours spent studying.

Equation of Simple Linear Regression:

$$ Y=\beta_0+\beta_1X+\varepsilon $$

Where:

- Y (Dependent Variable): The value to be predicted, such as examination marks.
- X (Independent Variable): The input used for prediction, such as study hours.
- β_0 (Intercept): The predicted value of Y when X is zero.
- β_1 (Slope): The change in the predicted value of Y for a one-unit increase in X.
- ε (Error Term): Represents the variation in the actual value that is not explained by the model.

Role of the Residual/Error Term:

A residual is the difference between the actual value and the predicted value.

Q4. How does Logistic Regression differ from Linear Regression? Write the logistic (sigmoid) function and explain why it is suitable for binary classification. What is the decision boundary in logistic regression?

Logistic Regression is a supervised learning algorithm mainly used for classification problems. Although its name includes regression, it predicts the probability that an observation belongs to a particular class.

Difference between Linear and Logistic Regression:

Basis	**Linear Regression**	**Logistic Regression**
Purpose	Regression	Classification
Output	Continuous numerical value	Probability between 0 and 1
Function	Linear equation	Sigmoid function
Example	Predicting house prices	Predicting whether a customer will leave a service

Logistic (Sigmoid) Function:

The logistic regression model first calculates a linear combination:

$$ z=\beta_0+\beta_1X_1+\cdots+\beta_nX_n $$

It then applies the sigmoid function:

$$ P(Y=1\mid X)=\frac{1}{1+e^{-z}} $$

Where:

- $P(Y=1\mid X)$ is the probability of belonging to class 1.
- e is the mathematical constant approximately equal to 2.718.
- z is the linear combination of input features and model coefficients.

Why is it suitable for binary classification?

The sigmoid function converts any real-valued input into a probability between 0 and 1. This makes it suitable for predicting outcomes such as yes/no, pass/fail, or spam/not spam.

Decision Boundary:

The decision boundary separates the two predicted classes. With a threshold of 0.5, the model predicts class 1 when the probability is at least 0.5 and class 0 otherwise.

For the standard logistic model, the boundary occurs when:

\[P(Y=1\mid X)=0.5 \]

Therefore, logistic regression is effective for predicting the probability and class of binary outcomes.

Q5. Describe how a Decision Tree works for classification. Explain Information Gain and Gini Impurity. Why can decision trees overfit the training data, and how can this problem be reduced?

A Decision Tree is a supervised learning algorithm that makes predictions through a tree-like structure of decisions. It divides the dataset into smaller groups based on feature values until a final class is predicted.

Working of a Decision Tree:

1. Root Node: Represents the complete training dataset.
2. Splitting: The algorithm selects a feature that best separates the classes.
3. Decision Nodes: Internal nodes represent conditions on selected features.
4. Leaf Nodes: Final nodes provide the predicted class.

For example, a decision tree may classify a loan application as approved or rejected based on income, credit score, and existing debt.

Information Gain:

Information Gain measures the reduction in uncertainty or entropy after splitting a dataset.

$$ IG=H(S)-\sum_{i=1}^{k}\frac{|S_i|}{|S|}H(S_i) $$

Where $H(S)$ is the entropy of the original dataset and $H(S_i)$ is the entropy of each resulting subset.

A higher Information Gain indicates a more useful split.

Gini Impurity:

Gini Impurity measures the probability of incorrectly classifying a randomly selected observation if it is labelled according to the class distribution of its group.

$$ Gini(S)=1-\sum_{i=1}^{k}p_i^2 $$

Here, p_i is the proportion of observations belonging to class i. A lower Gini value indicates a purer group.

Overfitting in Decision Trees:

Overfitting occurs when a tree becomes too complex and learns noise or minor details in the training data. It may achieve high training accuracy but perform poorly on new data.

Methods to Reduce Overfitting:

- Pruning: Removes unnecessary branches.
- Maximum Depth: Limits the depth of the tree.
- Minimum Samples: Requires a minimum number of observations for splitting or creating a leaf.
- Cross-Validation: Helps select appropriate model parameters.

Thus, a properly controlled decision tree provides interpretable predictions and better generalization.

Q6. What are Ensemble Methods? Differentiate between Bagging and Boosting with suitable examples of algorithms that use each technique. Why do ensemble methods generally perform better than a single model?

Ensemble Methods are machine learning techniques that combine the predictions of multiple models to produce a more accurate and reliable final prediction. These methods can reduce errors and improve the generalization ability of a model.

The two important ensemble techniques are Bagging and Boosting.

1. Bagging (Bootstrap Aggregating)

Bagging trains multiple models independently on different bootstrap samples of the training dataset. Their predictions are then combined using voting for classification or averaging for regression.

- Models can be trained in parallel.
- It primarily reduces variance.
- Example: Random Forest, which combines predictions from multiple decision trees.

2. Boosting

Boosting trains models sequentially, where each new model attempts to correct errors made by earlier models. The final prediction combines the contributions of the individual models.

- Models are trained one after another.
- It can improve accuracy by reducing bias and other prediction errors.
- Examples: AdaBoost, Gradient Boosting, and XGBoost.

Difference between Bagging and Boosting:

Basis	**Bagging**	**Boosting**
Training	Independent and parallel	Sequential
Main objective	Reduce variance	Improve weak learners
Data handling	Bootstrap samples	Focuses on errors or residuals
Example	Random Forest	AdaBoost, XGBoost

Why do Ensemble Methods Perform Better?

1. Improved Accuracy: Combining several models can produce better predictions than an individual model.
2. Reduced Variance: Bagging reduces sensitivity to changes in the training data.
3. Error Correction: Boosting focuses on mistakes made by previous models.
4. Better Generalization: Combining models can make predictions more reliable on unseen data.

However, ensemble methods do not always outperform a single model; their success depends on the quality and diversity of the models and the dataset.

Q7. Explain the working principle of Support Vector Machines (SVM) for binary classification. What are support vectors and the margin? Briefly describe the role of the kernel trick in SVM.

Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression. In binary classification, it aims to find an optimal decision boundary that separates observations belonging to two classes.

Working Principle of SVM:

1. Plot the Data: Represent the observations using their input features.
2. Find the Hyperplane: Identify a boundary that separates the two classes.
3. Maximize the Margin: Select the boundary that maximizes the distance from the nearest observations of both classes.
4. Classify New Data: Use the selected boundary to predict the class of unseen observations.

The decision boundary can be represented as:

$$ w^Tx+b=0 $$

Where w represents the weight vector, x represents the input features, and b is the bias.

Support Vectors:

Support vectors are the training observations closest to the decision boundary. They are important because they determine the position and orientation of the optimal hyperplane.

Margin:

The margin is the distance between the decision boundary and the nearest observations from the two classes. SVM attempts to maximize this margin to improve generalization.

For a standard hard-margin SVM, the margin width is:

$$ \text{Margin}=\frac{2}{\|w\|} $$

Kernel Trick:

Sometimes, data cannot be separated by a straight line or hyperplane in its original feature space. The kernel trick allows SVM to calculate relationships corresponding to a higher-dimensional feature space without explicitly transforming every observation.

Common kernels include:

- Linear Kernel
- Polynomial Kernel
- Radial Basis Function (RBF) Kernel

For example, an RBF kernel can help classify data with a nonlinear decision boundary.

Thus, SVM is useful for classification problems where a clear separation between classes can be achieved through an appropriate boundary and kernel.

Q8. List and briefly explain any four evaluation metrics used for classification problems. When is Accuracy not a good metric, and which metric would you prefer instead?

Classification evaluation metrics measure how effectively a model predicts the correct classes. The four commonly used metrics are Accuracy, Precision, Recall, and F1-Score.

First, consider the following terms:

- True Positive (TP): Positive cases correctly predicted as positive.
- True Negative (TN): Negative cases correctly predicted as negative.
- False Positive (FP): Negative cases incorrectly predicted as positive.
- False Negative (FN): Positive cases incorrectly predicted as negative.

1. Accuracy

Accuracy measures the proportion of all predictions that are correct.

$$ Accuracy=\frac{TP+TN}{TP+TN+FP+FN} $$

It is useful when the classes are reasonably balanced and the costs of different errors are similar.

2. Precision

Precision measures how many observations predicted as positive are actually positive.

$$ Precision=\frac{TP}{TP+FP} $$

It is important when false positives are costly, such as incorrectly marking a genuine email as spam.

3. Recall (Sensitivity)

Recall measures the proportion of actual positive observations correctly identified by the model.

$$ Recall=\frac{TP}{TP+FN} $$

It is important in medical screening, where missing a genuine positive case may be dangerous.

4. F1-Score

F1-Score is the harmonic mean of Precision and Recall.

\[F1=2\times\frac{Precision\times Recall}{Precision+Recall} \]

It is useful when both false positives and false negatives matter, especially with imbalanced classes.

When is Accuracy Not a Good Metric?

Accuracy can be misleading when the dataset is highly imbalanced. For example, if 95% of transactions are genuine and only 5% are fraudulent, a model that predicts every transaction as genuine achieves 95% accuracy but detects no fraud.

In such cases, Precision, Recall, F1-Score, or PR-AUC may be more informative. For fraud detection, Recall is important when catching as many fraudulent transactions as possible is the main goal, while Precision matters when false fraud alerts are costly. The choice depends on the application.

Q9. For regression problems, explain MSE, RMSE, MAE, and R-squared. Which metric is more sensitive to outliers and why?

Regression evaluation metrics measure the difference between actual and predicted numerical values. They help determine how well a regression model performs.

Let y_i represent the actual value, $\hat{y}_i$ represent the predicted value, $\bar{y}$ represent the mean of actual values, and n represent the number of observations.

(a) Mean Squared Error (MSE)

MSE calculates the average of the squared differences between actual and predicted values.

$$ MSE=\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2 $$

A lower MSE indicates smaller prediction errors. Because errors are squared, large errors receive much greater weight.

(b) Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

$$ RMSE=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2} $$

It expresses prediction error in the same units as the target variable, making it easier to interpret than MSE.

(c) Mean Absolute Error (MAE)

MAE calculates the average absolute difference between actual and predicted values.

$$ MAE=\frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i| $$

It is easy to interpret and treats errors proportionally to their absolute size. It is generally less sensitive to outliers than MSE and RMSE.

(d) R-squared (Coefficient of Determination)

R-squared measures how much of the variation in the target variable is explained by the regression model, relative to predicting the mean.

$$R^2=1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

An R^2 value of 1 indicates a perfect fit, while 0 indicates performance equivalent to predicting the sample mean under the usual evaluation setup. It can also be negative when the model performs worse than that baseline.

Comparison of Regression Metrics:

Metric	**Main Purpose**	**Interpretation**
MSE	Measures squared errors	Lower is better
RMSE	Measures error in original units	Lower is better
MAE	Measures average absolute error	Lower is better
R-squared	Measures explained variation	Generally higher is better

Which Metric is More Sensitive to Outliers?

MSE is highly sensitive to outliers because it squares each prediction error. For example, an error of 10 contributes 100 to the squared-error total, whereas an error of 2 contributes only 4. RMSE is also sensitive to outliers because it is derived from MSE.

MAE is generally more robust to extreme errors because it uses absolute values rather than squares.

Q10. Discuss any four real-world applications of Supervised Learning. For each application, mention the type of supervised learning problem and a suitable algorithm.

Supervised learning is widely used in real-world applications because it learns from labelled historical data to predict outcomes for new observations. Four important applications are explained below.

1. Email Spam Detection

Email spam detection identifies whether an incoming email is spam or legitimate.

- Type: Classification
- Input: Email text, sender details, and message features.
- Output: Spam or not spam.
- Suitable Algorithm: Naive Bayes or Logistic Regression.

The model learns from previously labelled emails and classifies new messages accordingly.

2. House Price Prediction

House price prediction estimates the selling price of a property using its characteristics.

- Type: Regression
- Input: Area, location, number of rooms, and property age.
- Output: Predicted house price.
- Suitable Algorithm: Linear Regression or Random Forest Regressor.

The model learns from historical property data to estimate prices for new properties.

3. Medical Disease Diagnosis

Supervised learning can help identify whether a patient is likely to have a particular disease based on medical data.

- Type: Classification

- Input: Symptoms, test results, and relevant patient measurements.
- Output: Disease-positive or disease-negative prediction.
- Suitable Algorithm: Logistic Regression, Decision Tree, or SVM.

The model learns from labelled medical records and supports healthcare professionals in identifying potential cases. Its predictions require appropriate clinical validation.

4. Sales and Demand Forecasting

Businesses use supervised learning to estimate future sales or product demand.

- Type: Regression
- Input: Historical sales, prices, promotions, and seasonal factors.
- Output: Predicted sales quantity or revenue.
- Suitable Algorithm: Linear Regression, Random Forest Regressor, or Gradient Boosting Regressor.

The model helps businesses plan inventory, allocate resources, and make purchasing decisions.

Summary Table:

Application	**Problem Type**	**Suitable Algorithm**
Email spam detection	Classification	Naive Bayes
House price prediction	Regression	Linear Regression
Medical disease diagnosis	Classification	Logistic Regression
Sales forecasting	Regression	Random Forest Regressor

In conclusion, supervised learning is useful for both classification and regression tasks. It helps automate decision-making, identify patterns in historical data, and make predictions that support real-world business, healthcare, and technology applications.